CS341 · COMPUTER GRAPHICS

Chapter 3 Color Models

Midterm summary

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What's inside

01 Physical properties of light

02 Psychological properties of light

03 Color definitions (asked as fill-in-the-blank)

04 Additive and subtractive mixing

05 Human color vision

06 CIE and the chromaticity diagram

07 RGB

08 CMY and CMYK

09 YIQ

10 HSV

11 HLS

01 Physical properties of light

- Physically, color is electromagnetic radiation within a narrow frequency band.
- Visible light: 380 to 750 nm.
- $1\ \text{nm} = 10^{-9}\ \text{m}$ and $1\ \text{Å} = 10^{-10}\ \text{m}$, so $1\ \text{nm} = 10\ \text{Å}$.
- The relation: $c = \lambda \cdot f$, i.e. $\lambda = \frac{c}{f}$.

	Red	Violet
Frequency	Low, $4.3 \times 10^{14}\ \text{Hz}$	High, $7.5 \times 10^{14}\ \text{Hz}$
Wavelength	Long, ~700 nm	Short, ~400 nm

From low to high frequency: red, orange, yellow, green, blue, violet.

02 Psychological properties of light

Property	Meaning
Color (hue)	The dominant frequency
Brightness	The total light energy, quantified as luminance
Purity / Saturation	How close the light is to a pure spectral color such as red
Chromaticity	Hue and purity together
Luminance	The quality of emitting or reflecting light

03 Color definitions (asked as fill-in-the-blank)

Term	Definition
Color model	Any method for explaining the properties of color in some context. No single model explains everything.
Primary colors	The basic colors used to produce the others; they cannot be made from each other.
Color gamut	The set of all colors we can produce by mixing the primary colors.
Complementary colors	Two colors that combine to produce white.
Shade	Add a **black** pigment to the pure color.
Tint	Add a **white** pigment, making it lighter.
Tone	Add **both white and black** pigments.
Grays	Black pigment plus white pigment.

Asked in the midterm

The definitions of color gamut, tint and tone came as fill-in-the-blank, word for word.

04 Additive and subtractive mixing

Additive (RGB)

Light is added. Pure colors sit close together and mix on the retina. The technique of displays.

Red + Green = Yellow
Red + Blue = Magenta
Green + Blue = Cyan
Red + Green + Blue = White
None = Black

Subtractive (CMY)

Pigments (ink) are mixed. Each pigment absorbs part of the light. The technique of printers.

Magenta + Yellow = Red
Magenta + Cyan = Blue
Cyan + Yellow = Green
Magenta + Yellow + Cyan = Black
None = White

The slide's example: yellow absorbs blue, so only red and green reach the eye, and together they look yellow.

Asked in the midterm

M + Y + C = White → **False** (black)
Yellow + Magenta = **Red** Magenta + Cyan = **Blue** Green + Blue = **Cyan**

05 Human color vision

- The retina contains cones (7–8 million) and rods (100–120 million).
- Color perception is the "sum" of the light's energy spectrum, the object's reflection spectrum and the eye's response spectrum.
- Trichromacy: the color we see depends on signals from three cone types (L, M, S), so it can be mapped to three colors R, G, B. The three numbers are the tristimulus values.

- The eye perceives about 382,000 colors, so a classification system with three coordinates is needed.

The six models: CIE, RGB, CMY/CMYK, YIQ, HSV, HLS.

06 CIE and the chromaticity diagram

CIE (International Commission on Illumination, 1931) uses three imaginary primaries `X, Y, Z` (nonexistent, super-saturated).

$$S = A \cdot X + B \cdot Y + C \cdot Z$$

$$x = \frac{A}{A + B + C} \qquad y = \frac{B}{A + B + C} \qquad z = \frac{C}{A + B + C}$$

$$x + y + z = 1$$

Plotting `x` and `y` for the visible spectrum gives a tongue-shaped curve called the CIE chromaticity diagram:

- Points on the curve are pure spectral colors.
- The dotted line joining red and violet is the purple line; it is not part of the spectrum.
- Interior points are all visible color combinations. Point `C` is white.
- Luminance is not shown because of normalization.

Chromaticity diagram is useful for:
1. Comparing color gamuts for different sets of primaries.
2. Identifying complementary colors.
3. Determining purity and dominant wavelength for a given color.

Concept	On the diagram
Color gamut	Two colors: the line between them. Three colors: the triangle (only colors inside or on its edges).
Complementary	Two points on opposite sides of C, collinear with it. Their distances to C set the amounts needed for white.
Dominant wavelength	Draw a line from C through the color to the curve; the point Cs is the dominant wavelength. If it hits the purple line, take its complement on the curve (Csp).
Purity	$\text{purity} = \frac{dc1}{dcs}$ (distance of the color from C ÷ distance of Cs from C)

No choice of three primaries covers all visible colors. RGB is the natural choice because it covers the largest part of the horseshoe. A monitor's gamut differs from a printer's.

07 RGB

- An **additive** model used by displays: CRT, LCD, plasma and LED.
- The screen emits light and its surface is black, so the origin `(0, 0, 0)` is black.
- Colors are not absolute; they depend on each device's settings.

```
C = r·R + g·G + b·B,        r, g, b ∈ [0, 1]
```

```
R = (1,0,0)   G = (0,1,0)   B = (0,0,1)
```

```
R + G = (1,1,0) = Yellow     R + B = (1,0,1) = Magenta
```

```
G + B = (0,1,1) = Cyan       R + G + B = (1,1,1) = White
```

```
Grays = (x, x, x)            e.g. (0.5, 0.5, 0.5)
```

The RGB cube: black at the origin, white at the opposite corner, grays on the diagonal between them. One bit per primary gives 8 colors; 8 bits per primary gives 16 million colors (true color).

08 CMY and CMYK

- A **subtractive** model, the complement of RGB, used by printers and plotters.
- The paper is white, so the origin `(0, 0, 0)` is white and `(1, 1, 1)` is black (the reverse of RGB).
- Cyan absorbs red, magenta absorbs green, yellow absorbs blue.
- Example: white light reflected from cyan ink contains only green and blue; red is absorbed.

```
[C, M, Y] = [1, 1, 1] − [R, G, B]        (RGB → CMY)
```

```
[R, G, B] = [1, 1, 1] − [C, M, Y]        (CMY → RGB)
```

```
Orange (1, 0.5, 0)  →  CMY = (0, 0.5, 1)
```

CMYK adds K (black, "key") because mixing C, M and Y does not produce a fully saturated black. It is the printing model.

09 YIQ

- The NTSC encoding for US color-TV broadcasting. An RGB monitor needs three separate signals; a TV uses one composite signal.
- `Y` is luminance (brightness); `I, Q` carry chromaticity (hue and purity).
- Black-and-white TVs use only `Y`, so the system is backward compatible.

```
Y = 0.299 R + 0.587 G + 0.114 B
```

```
I = 0.596 R − 0.275 G − 0.321 B
```

```
Q = 0.212 R − 0.528 G + 0.311 B
```

```
White (1,1,1):  Y = 0.299 + 0.587 + 0.114 = 1
```

```
Yellow (1,1,0): Y = 0.299 + 0.587 = 0.886
```

10 HSV

RGB, CMY and YIQ are hardware-oriented. HSV and HLS are user-oriented. HSV is also called HSB (B = brightness).

The shape is a hexcone. Looking at the RGB cube along its diagonal from white to black gives a hexagon, which forms the top of the hexcone.

Parameter	Range	Meaning
H (hue)	`0° – 360°`	Angle around the vertical axis: 0° red, 60° yellow, 120° green, 180° cyan, 240° blue, 300° magenta
S (saturation)	`0 – 1`	Distance from the center = purity. 1 is pure; lower values move toward gray
V (value)	`0 – 1`	Along the vertical axis: 0 at the apex (black), 1 at the top (white)

```
Pure hues: S = 1, V = 1
```

```
White = (–, 0, 1)    Black = (–, 0, 0)
```

```
Red = (0, 1, 1)      Yellow = (60, 1, 1)   …   Magenta = (300, 1, 1)
```

```
Dark blue:  H = 240°, S = 1.0, V = 0.4
```

```
Light blue: H = 240°, S = 0.3, V = 1.0
```

Concept	In HSV	Effect
Tint	`V = 1, 0 ≤ S ≤ 1`	Saturation reduced
Shade	`S = 1, 0 ≤ V ≤ 1`	Lightness reduced
Tone	Both reduced	Saturation and lightness reduced

The eye distinguishes 128 hues, 130 tints and 23 shades of yellow (16 of blue): `128 × 130 × 23 = 382,720` colors. A true-color system has 16 million colors.

11 HLS

- Hue, lightness, saturation. The shape is a double cone whose base is the HSV hexagon.
- `L = 0` is black and `L = 1` white; grays lie on the L axis and pure colors on the `L = 0.5` plane.

```
(h, l, s):  h ∈ [0°, 360°),  l, s ∈ [0, 1]
```

```
White = (–, 1, 0)     Black = (–, 0, 0)
```

```
Red = (0, 0.5, 1)     Yellow = (60, 0.5, 1)
```

	HSV	HLS
Shape	Hexcone	Double cone
Pure color	`S = 1, V = 1`	`S = 1, L = 0.5`
White	`(–, 0, 1)`	`(–, 1, 0)`

Paint programs show more than one model in their color picker (Hue/Sat/Lum and Red/Green/Blue).